

DATABRICKS · HEALTHCARE / ICU OPERATIONS

ICU Step-Down Readiness

Turning scarce ICU capacity into a decision advantage — an AI-assisted way to see, every shift, which patients are most ready to safely leave intensive care.

Lakeflow

Unity Catalog

Lakebase

Mosaic AI + Foundation Models

Genie

Databricks Apps

THE PROBLEM

A blocked ICU bed is a clinical and financial bottleneck

70–90%

Typical ICU occupancy
— one blocked bed
delays a critical
admission

Manual

Step-down decisions
are subjective and vary
by provider and shift

**"To be
safe"**

Clinicians hold patients
longer than needed,
lengthening stays

The question every intensivist asks on rounds — *“who can safely move to step-down today?”*
— has no consistent, data-driven answer.

WHY IT MATTERS BEYOND THE UNIT

ICU length of stay is a margin, benchmarking, and market-share lever

Payer & margin risk

Under DRG-fixed and value-based contracts, every avoidable ICU day is **unreimbursed cost**, and ICU bounce-backs feed readmission penalties and quality scores (CMS HRRP, star ratings) that payers price on.

Peer benchmarking

ICU LOS, occupancy, and throughput are ranked against peer hospitals (Vizient, CMS, Leapfrog). Being an **LOS outlier** is a board- and payer-visible signal of inefficiency.

Competitive position & growth

A full ICU means **diverted ambulances, declined tertiary referrals, and postponed elective surgery** — lost revenue, lost referral relationships, and lost market share. Freed capacity is growth capacity.

The intensivist's daily step-down call, aggregated across the unit, is what a **CFO, CMO, and market strategist** see as reimbursement exposure, a benchmark percentile, and the ability to say yes to the next transfer.

THE OUTCOME WE SELL

Free ICU capacity without compromising safety

**10–
15%**

Target
reduction
in ICU
length of
stay

**\$2–
5k**

Saved per
accelerated
discharge-
day

+beds

More ICU
capacity for
emergent
admissions

Consistency

Evidence-informed
decisions across shifts

For a 30-bed ICU, a 0.8-day average LOS reduction is on the order of **hundreds of additional patient-days of capacity per year**.

WHAT WE BUILT

One governed journey — raw data to a clinician's screen



Integrated, not siloed: the same governed gold tables feed the operational app, the model, and natural-language querying — one lineage, one source of truth.

THE PRODUCT

A ranked ICU census the care team acts on

- **Census dashboard** — every current patient ranked by a relative readiness index, color-banded Ready / Borderline / Not-ready.
- **Patient detail** — the factors driving each recommendation, vital-sign trends, and a plain-language **AI clinical summary**.
- **Analytics** — capacity and outcome trends, with honest model performance.

Live: [icu-step-down-...
databricksapps.com](#)

Example — ranked census

- **92** · Ready · off pressors 36h, GCS 15
- **78** · Ready · extubated, stable VS
- **54** · Borderline · lactate ↑
- **18** · Not ready · on vent + pressors

FOR THE CMIO & ICU MEDICAL DIRECTOR

Built to be trusted, governed, and operated

Governed by design

Unity Catalog across the medallion — PII classification tags on identifiers, least-privilege grants, and full bronze→gold lineage.

Operationally served

Gold features synced to Lakebase (Postgres) for low-latency app reads; model served on Mosaic AI; narrative via the Foundation Model API.

Reproducible

Everything deploys via Databricks Asset Bundles; 117 backend + 32 frontend + 17 pipeline-integration tests.

Safe by framing

De-identified public data; a relative decision-support index, not an automated discharge gate.

MODEL HONESTY

Decision support, not a discharge switch

AUC 0.66

Honest, modest
discrimination on the
retrospective feature
set

Index, not %

Relative readiness
rank within the current
census — never a
"probability of safe
discharge"

Explainable

SHAP factors in clinical
language; known risks
always shown as risks

The roadmap to stronger discrimination: richer real-time features, calibration, and prospective validation with the clinical team.

THE ASK

From prototype to a unit pilot

- **VP of Clinical Operations / CMO** — measured on ICU length of stay, bed occupancy, and emergent-admission wait times: sponsor a 60–90 day pilot in one ICU, scored on LOS reduction with the safe-transfer rate held or improved.
- **ICU Medical Director** — measured on safe-transfer rate and 48–72h ICU bounce-back: co-define the readiness features and validate the index against your step-down criteria.
- **CMIO / VP Data & Analytics** — measured on time-to-value and governed reuse: point the same pipeline at your EHR feed — the architecture is unchanged.

Every ask maps to a number already on that leader's own scorecard. The capability is live today on Databricks; the next step is your data and your clinicians.